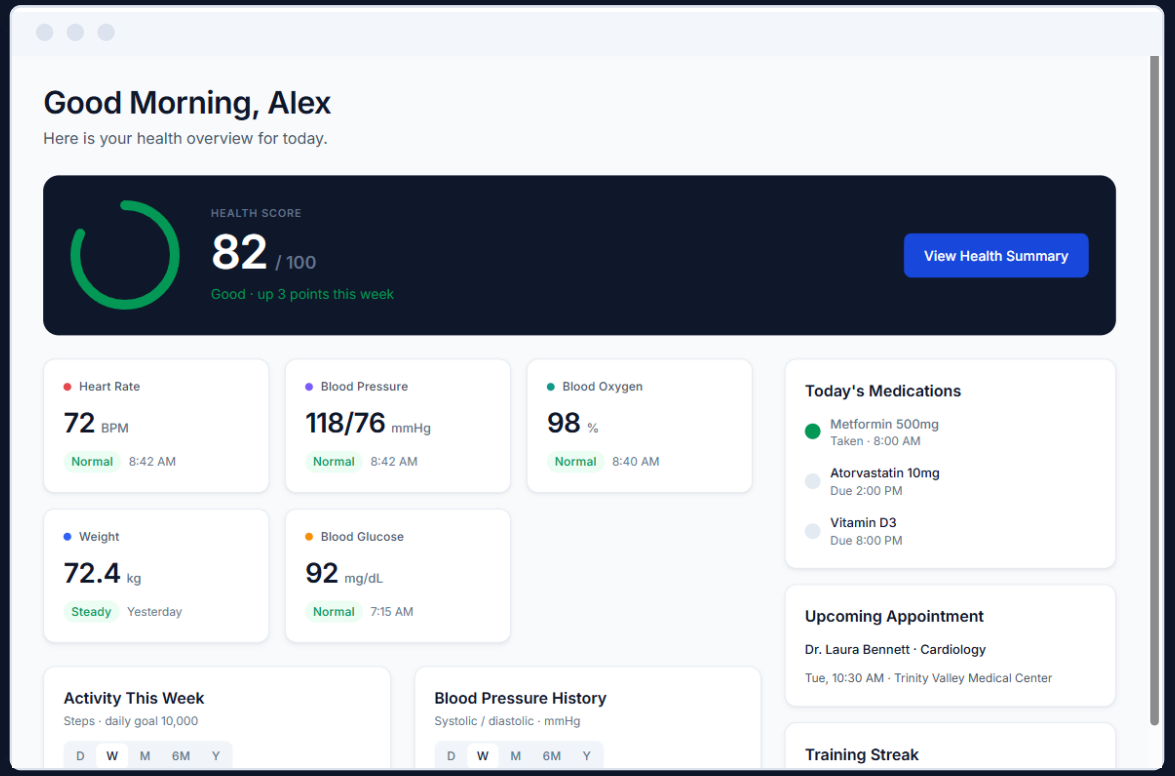
VitalLink

A consumer health and fitness web portal

Final Project Presentation

The University of Texas at Arlington, Group 2

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Agenda

What we will cover today

01	**Problem domain**	The problem, and the principles we set
02	**The application**	What each feature does and how it works
03	**Tools and technologies**	Architecture, stack and testing
04	**Use of generative AI**	How much we used it, and what we learned
05	**Live demonstration**	The working system, end to end
06	**Limitations and questions**	What comes next

Problem Domain

Personal health data is scattered across devices, apps and paper

Devices

Steps and heart rate stay in a phone or watch.

Separate apps

Workouts and meals are tracked in different places.

Email and paper

Lab results and documents are hard to find when needed.

Memory

Medications and appointments are easy to miss.

Sharing

Showing a doctor or relative is usually all or nothing.

Visibility

People rarely see who looked at their data, or when.

Project Scope

Three principles guided every decision

01

Consumers only

No hospital or provider system. A doctor who receives shared data uses an ordinary account.

02

Free to operate

Hosting, database, email and automation all run on free plans.

03

Not a medical device

It presents a person's data and never diagnoses. The FTC Health Breach Notification Rule applies, not HIPAA.

Dashboard

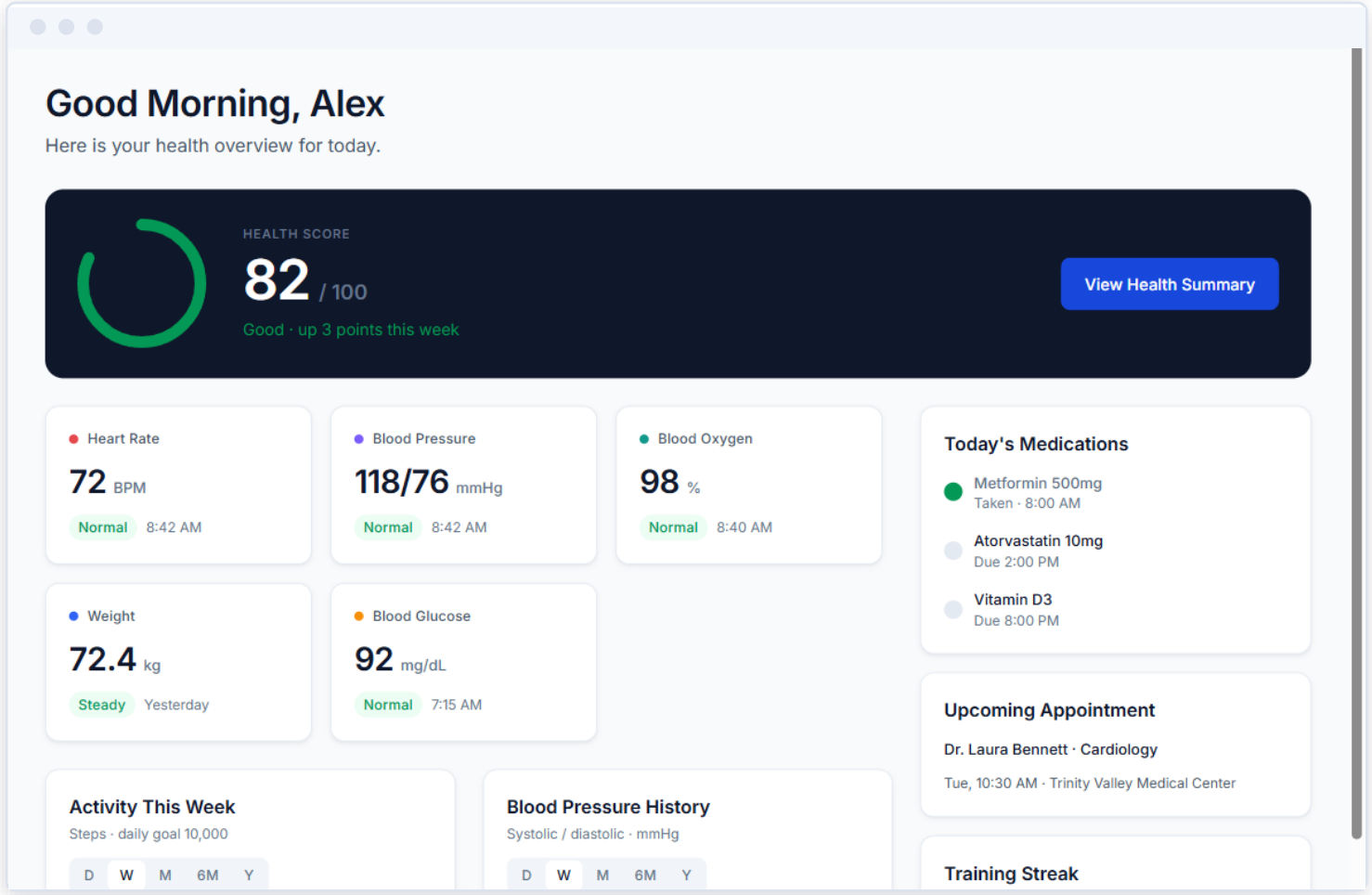
A single daily view of health, medications and activity

What it does

Shows the health score, today's vitals, medications, the next appointment and activity charts.

How it works

- Readings are stored as 5-minute blocks, then totalled by day in the user's time zone.
- The score compares last night's sleep, resting heart rate and HRV with the user's goal and 30-day baseline.



My Health

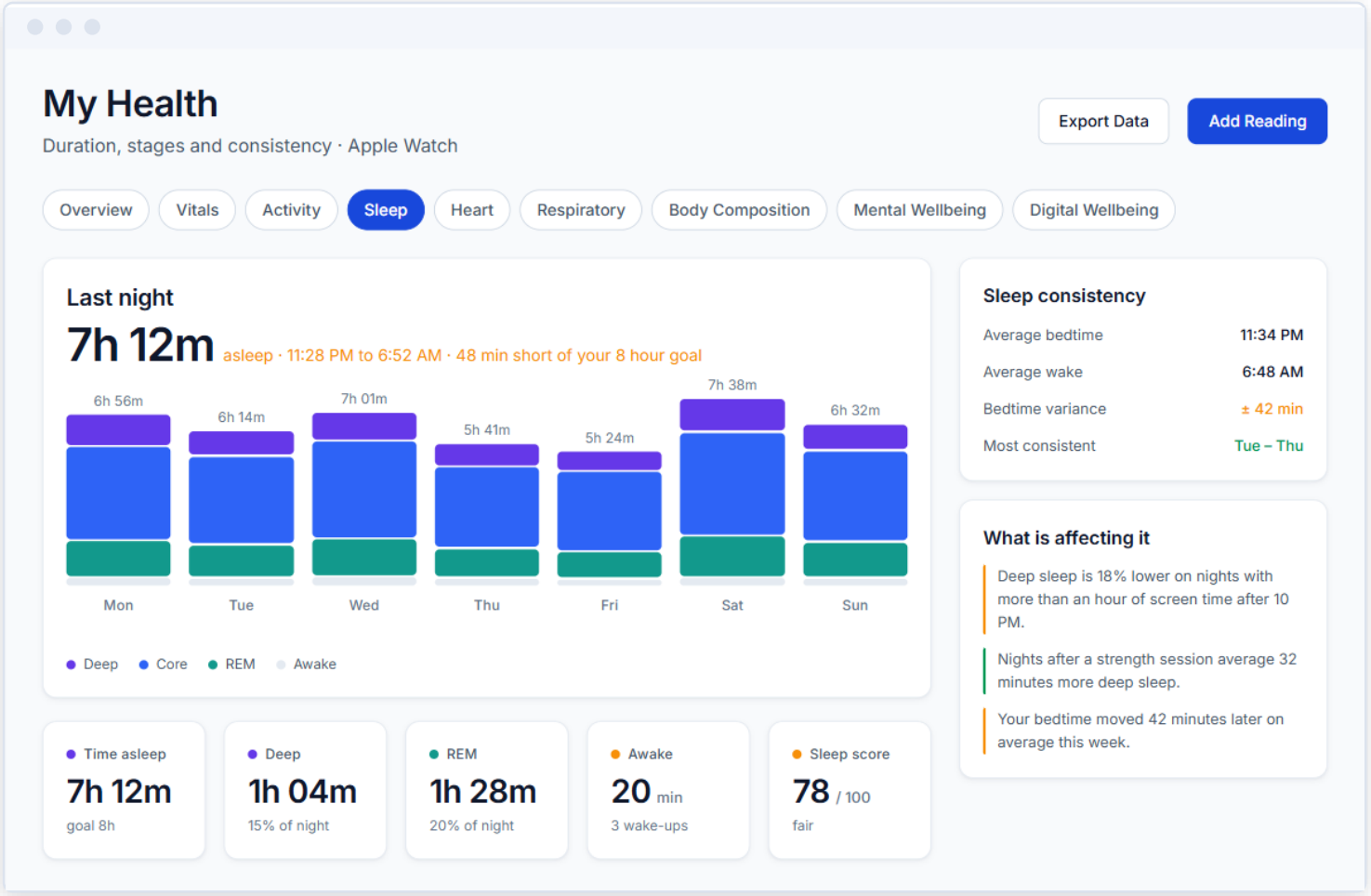
Nine detailed views of a person's health data

What it does

Overview, vitals, activity, sleep, heart, respiratory, body composition, mental wellbeing and digital wellbeing.

How it works

- Each view is calculated when it is opened, so it is never out of date.
- Anything that is not measured is left out, not estimated.
- Insights compare a person with their own history.



Data Import

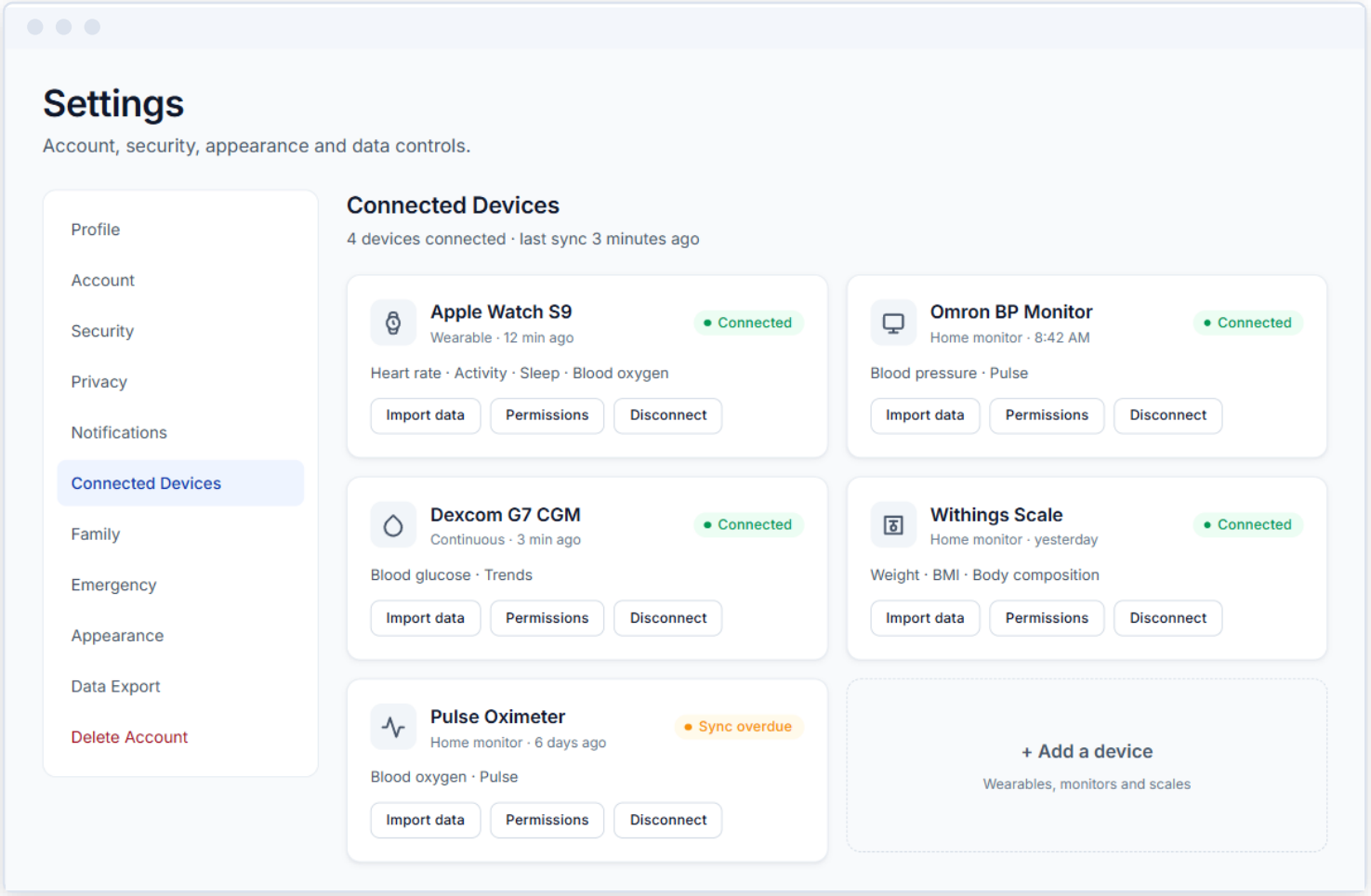
Three ways to bring health data into VitalLink

What it does

Import an Apple Health export, pair a Bluetooth device, or enter a reading by hand.

How it works

- Apple's export can be very large, so it is read as a stream, not all at once.
- iPhone and Watch count the same steps, so each block uses one source.
- Importing again updates readings and does not duplicate them.



Workouts

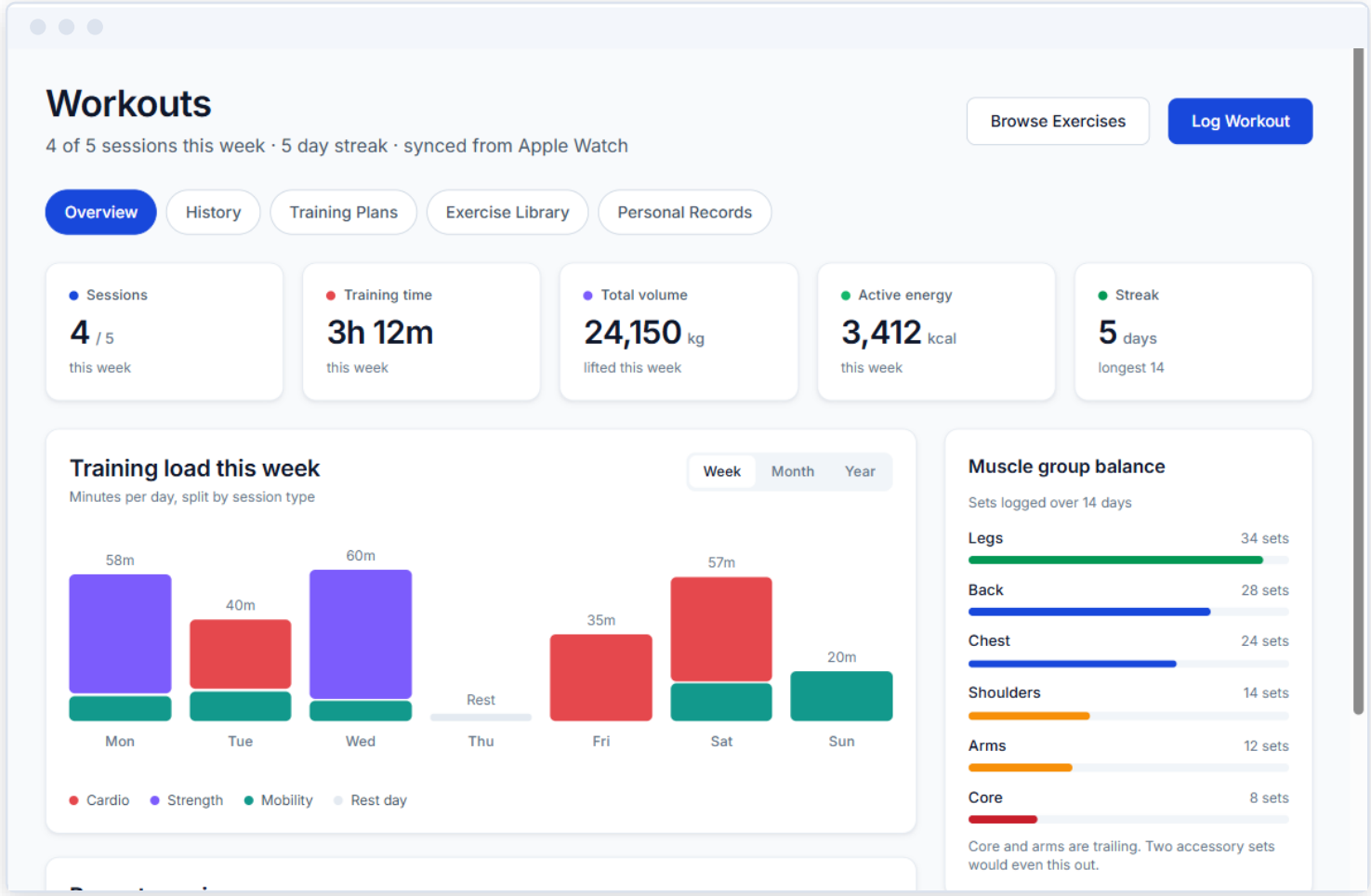
Training logs, plans and progress in one place

What it does

Log workouts and sets, follow a training plan, browse exercises and track personal records.

How it works

- Weekly load, muscle balance and streaks are calculated from logged sessions.
- One plan is active at a time. Starting a new plan archives the previous one.



Nutrition

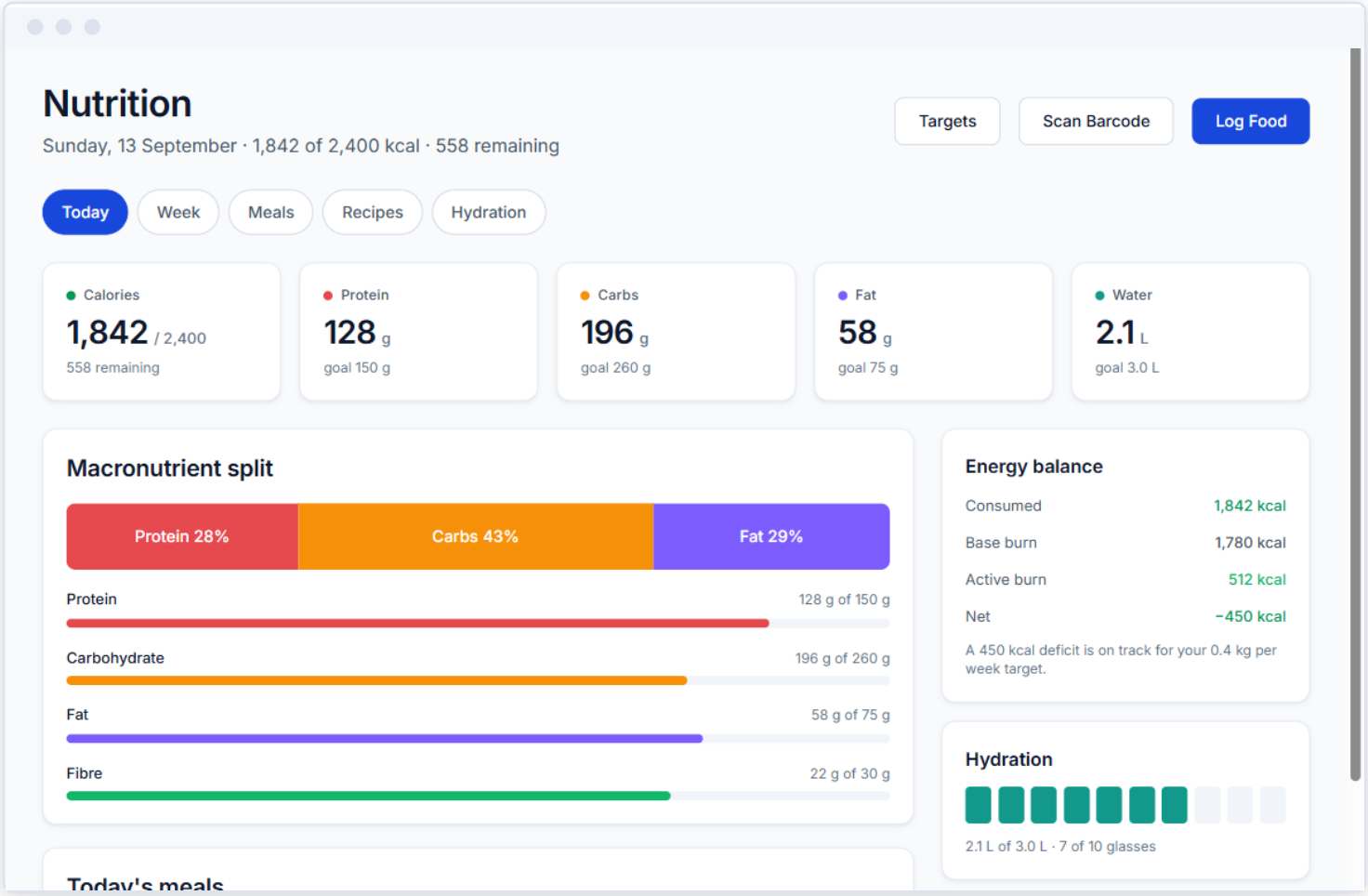
Meals, recipes, barcodes and hydration

What it does

Log meals, scan a barcode, save recipes, track water and set personal targets.

How it works

- Barcodes are looked up in Open Food Facts. Only the barcode is sent.
- Logging copies the nutrition values, so editing a recipe never changes past meals.



Goals

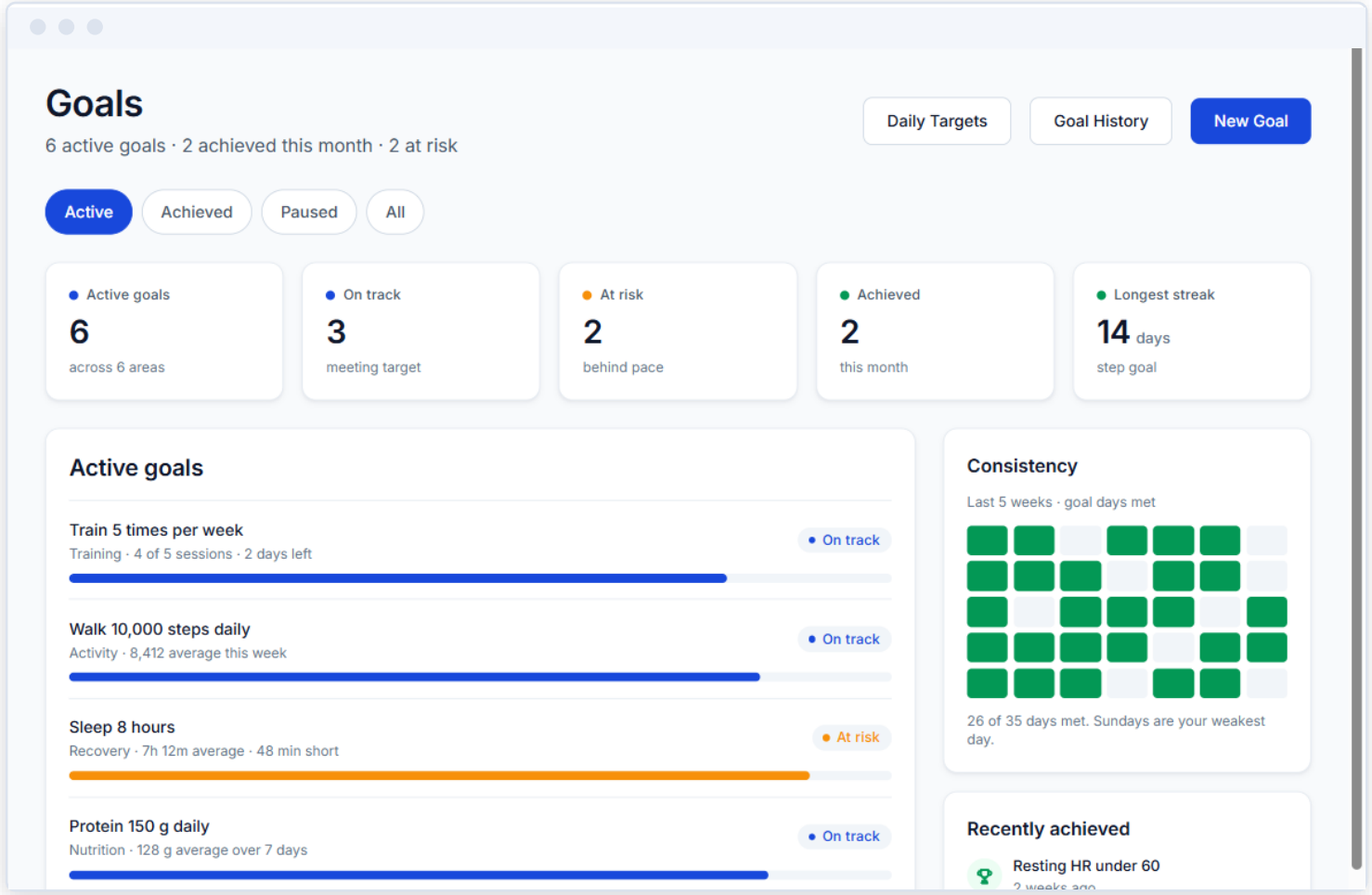
Clear progress against daily, weekly and target goals

What it does

Set goals, then follow progress, streaks and consistency over time.

How it works

- Daily goals use the average of the last 7 days with data. 85% of target counts as on track.
- Weekly goals check pace against the days gone.
- Status is calculated on each read, so it cannot go stale.



Medical Records and Lab Results

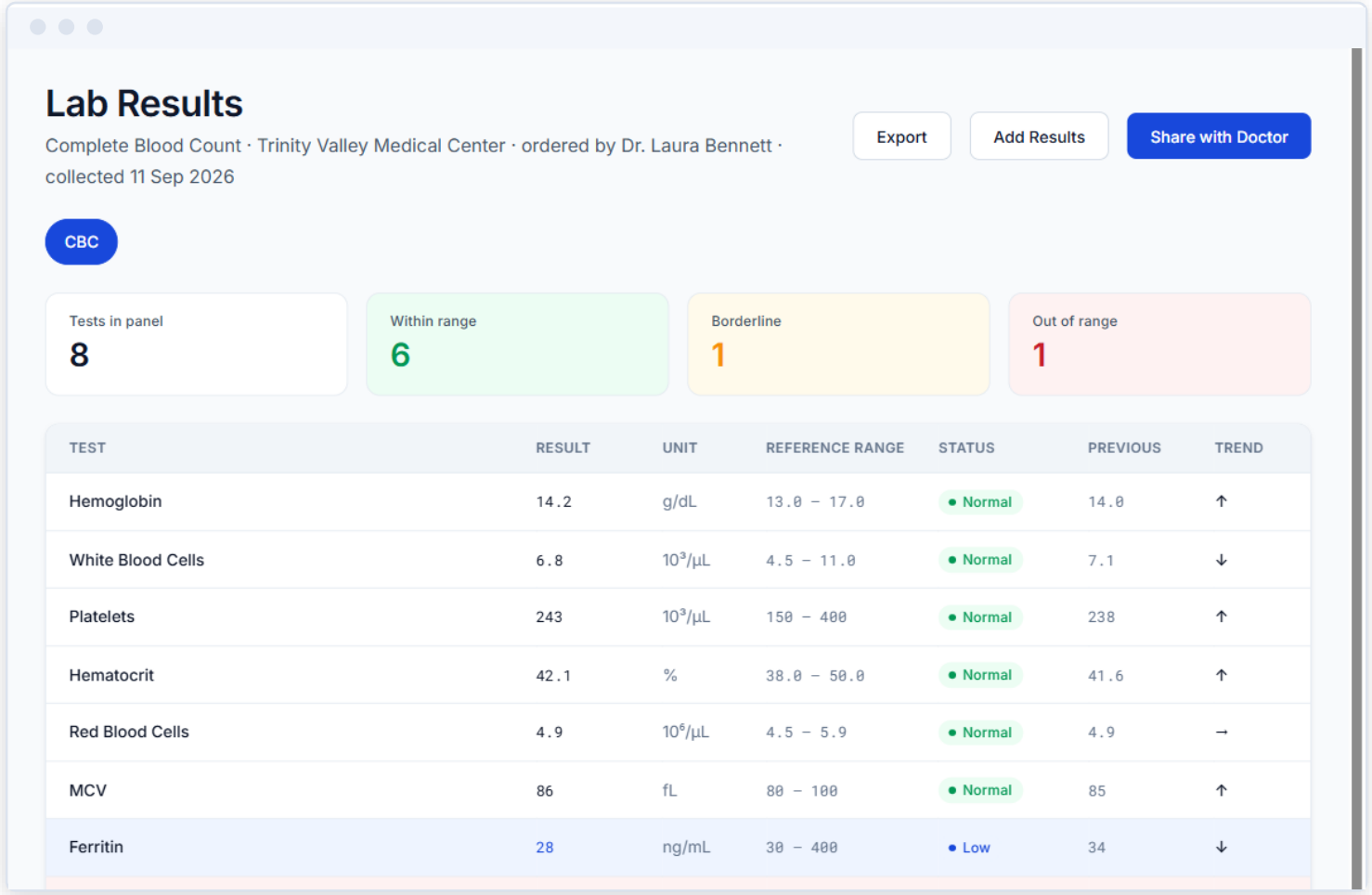
Documents and results that stay under the user's control

What it does

Upload documents, enter lab results, spot values out of range and export to CSV.

How it works

- Files stay in private storage and open only through short-lived signed links.
- The server checks a file's real type after upload.
- Each result is compared with its reference range and previous value.



Medications

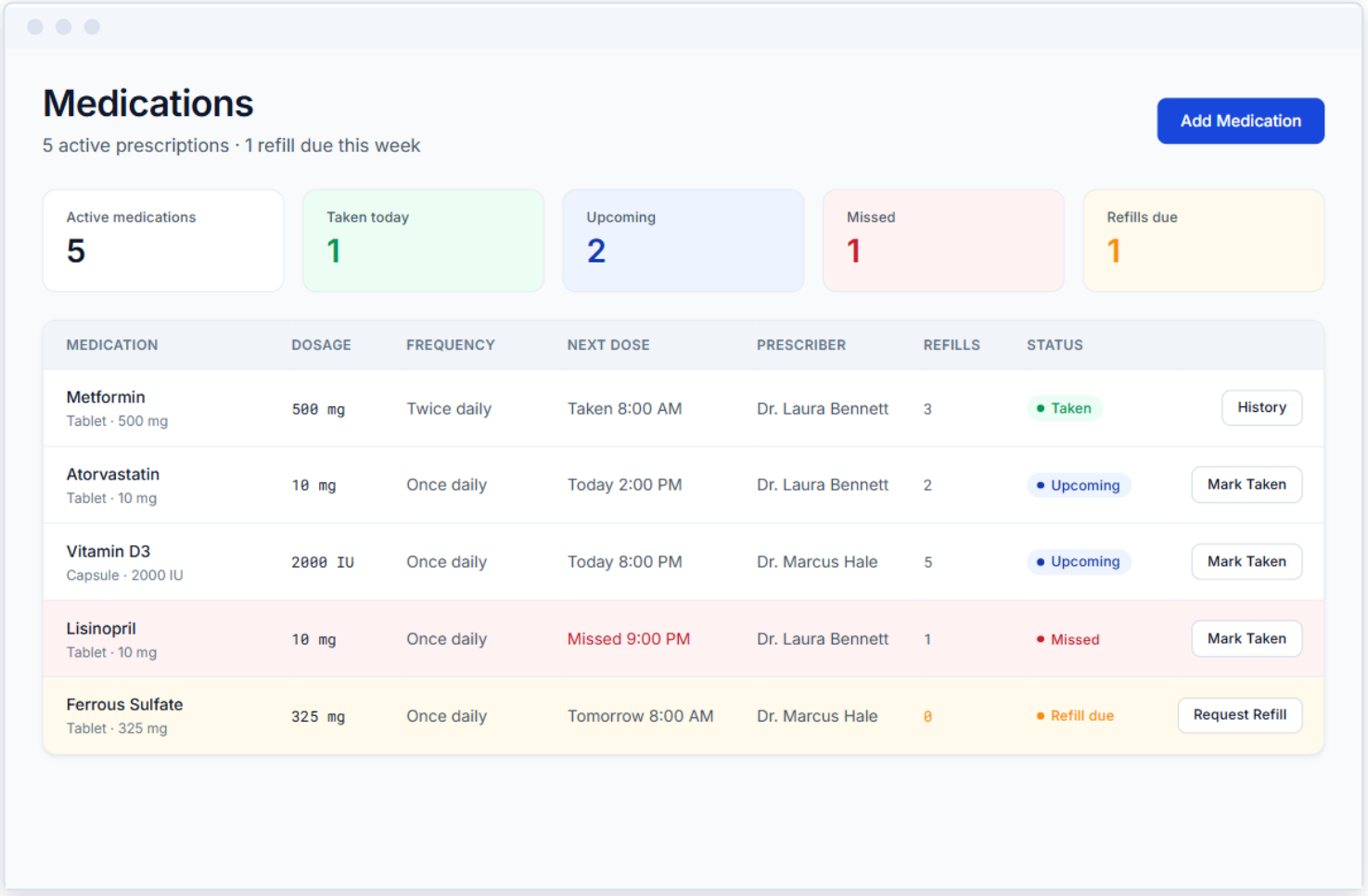
Doses, history and refills

What it does

See what is due, mark a dose as taken, undo it, review history and record refills.

How it works

- A dose stays due for 2 hours, then counts as missed.
- Marking the same dose twice records it once.
- A nightly job creates notifications for missed doses, refills and the next day's visits.



Appointments and Doctor Search

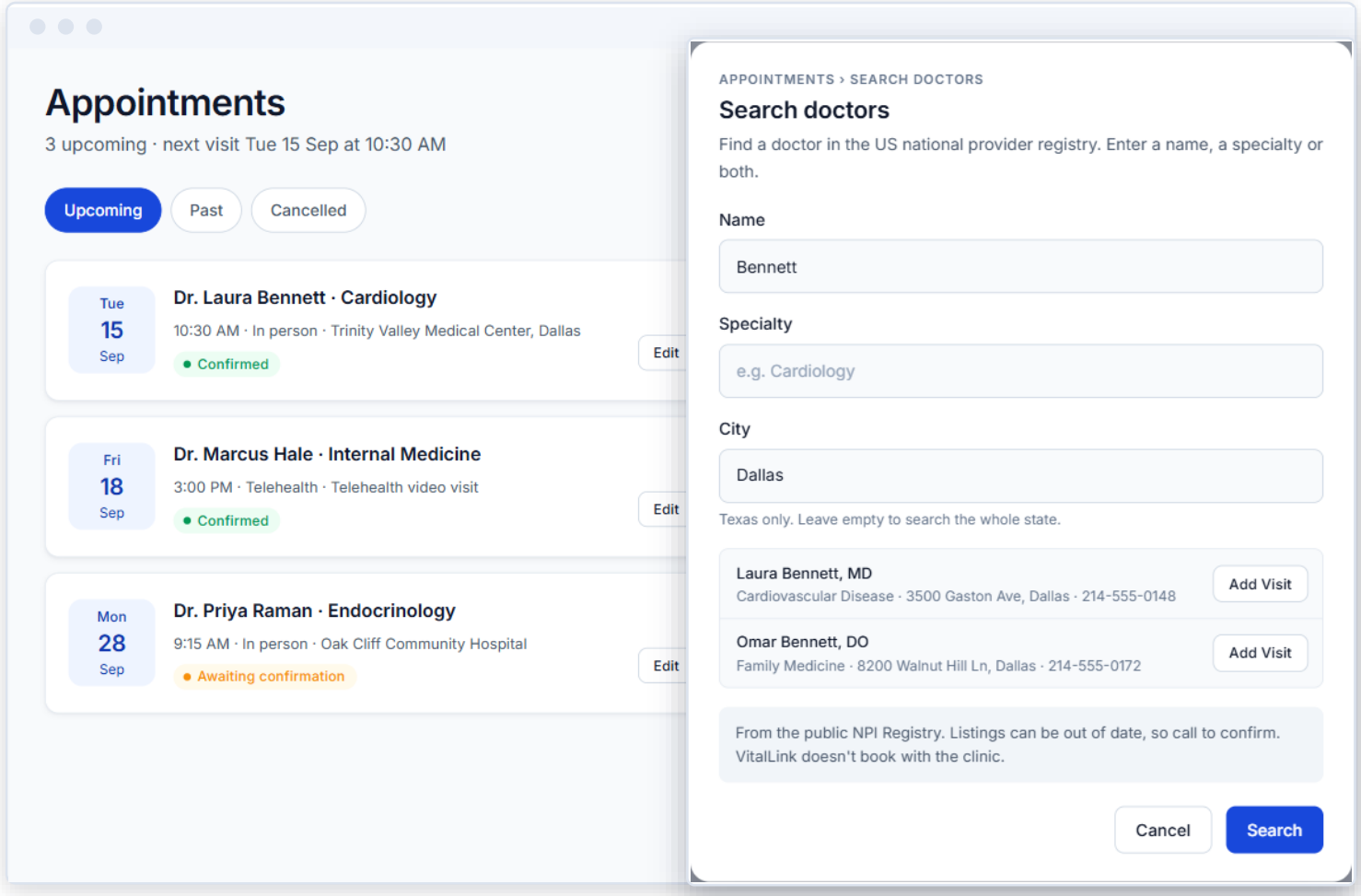
Track visits and find doctors from a public registry

What it does

Track visits and get directions. Search Texas doctors by name, specialty and city.

How it works

- Search uses the government's public NPI Registry. Only the search words are sent.
- Add Visit fills in the booking form.
- VitalLink does not book with the clinic. It keeps the calendar and reminders.



Emergency Medical ID

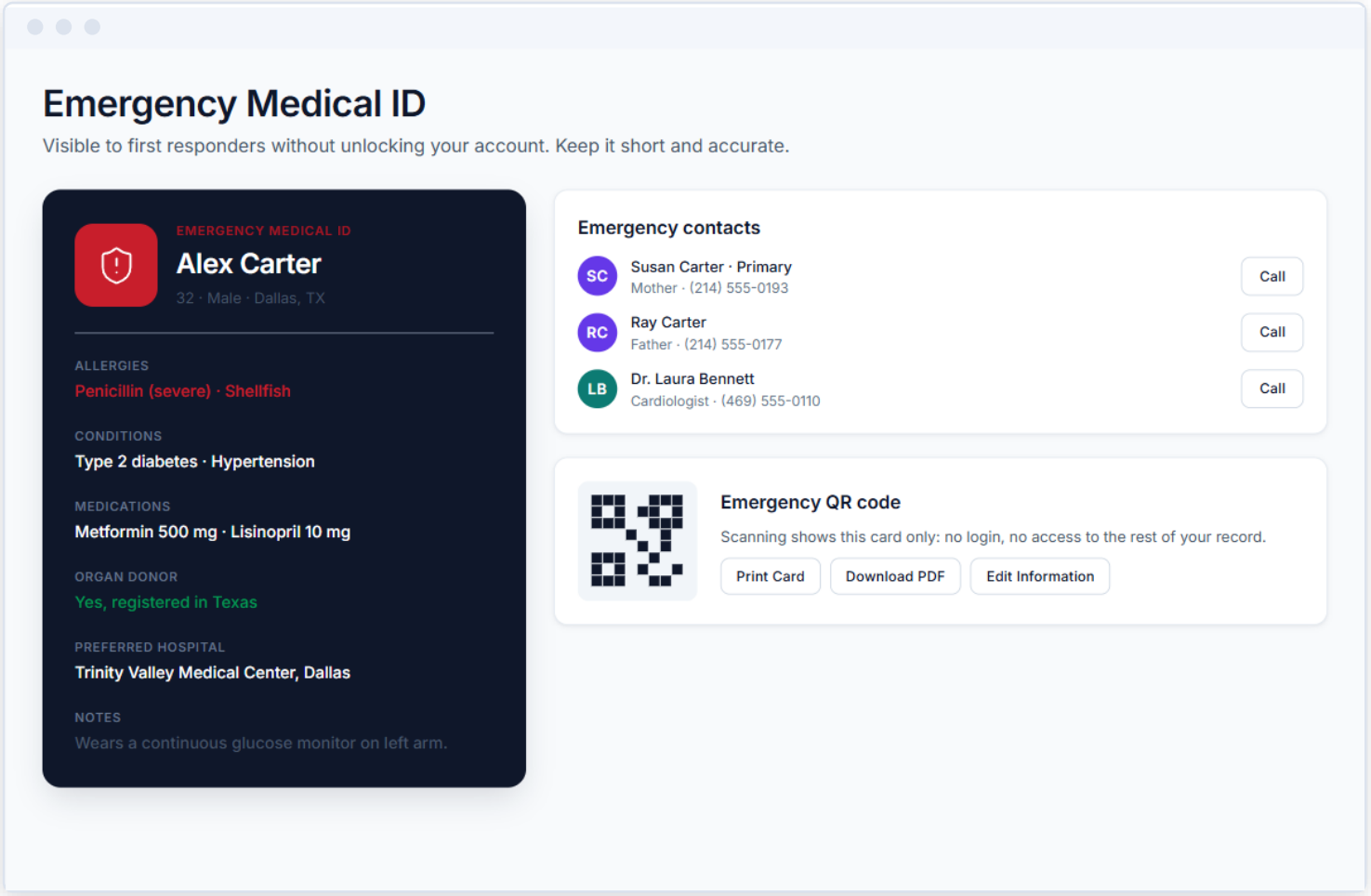
Critical details on one card

What it does

One card with allergies, conditions, medications and emergency contacts, ready to print.

How it works

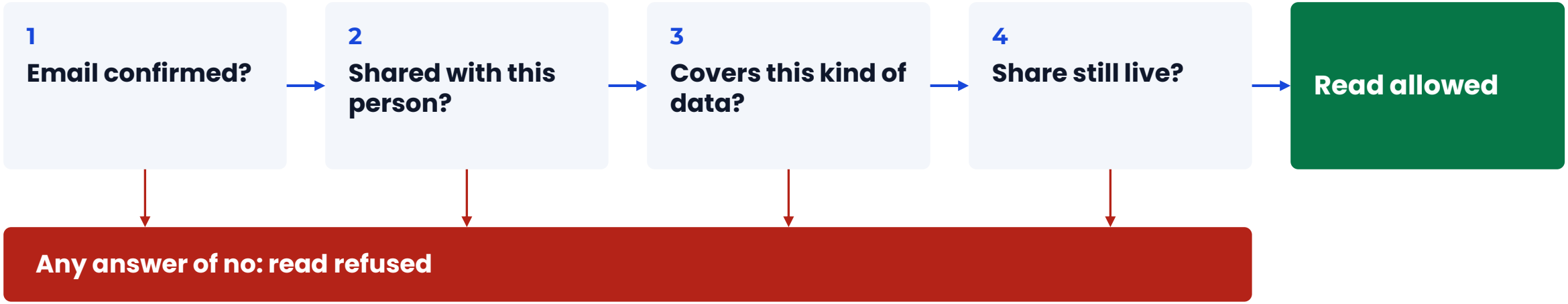
- The card is built from data already entered, so nothing is typed twice.
- Up to five contacts, with one marked primary.
- It can be printed or saved as a PDF from the browser.



Data Sharing and Consent

Users choose who sees which data and for how long, and can stop at any time

Every read of another person's data passes four checks



Limited by kind and time

Vitals, activity, sleep, nutrition or records, with an end date.

Fully logged

Allowed and refused reads go into a log the owner can read.

No account probing

Strangers cannot tell whether an account exists.

Accounts and Security

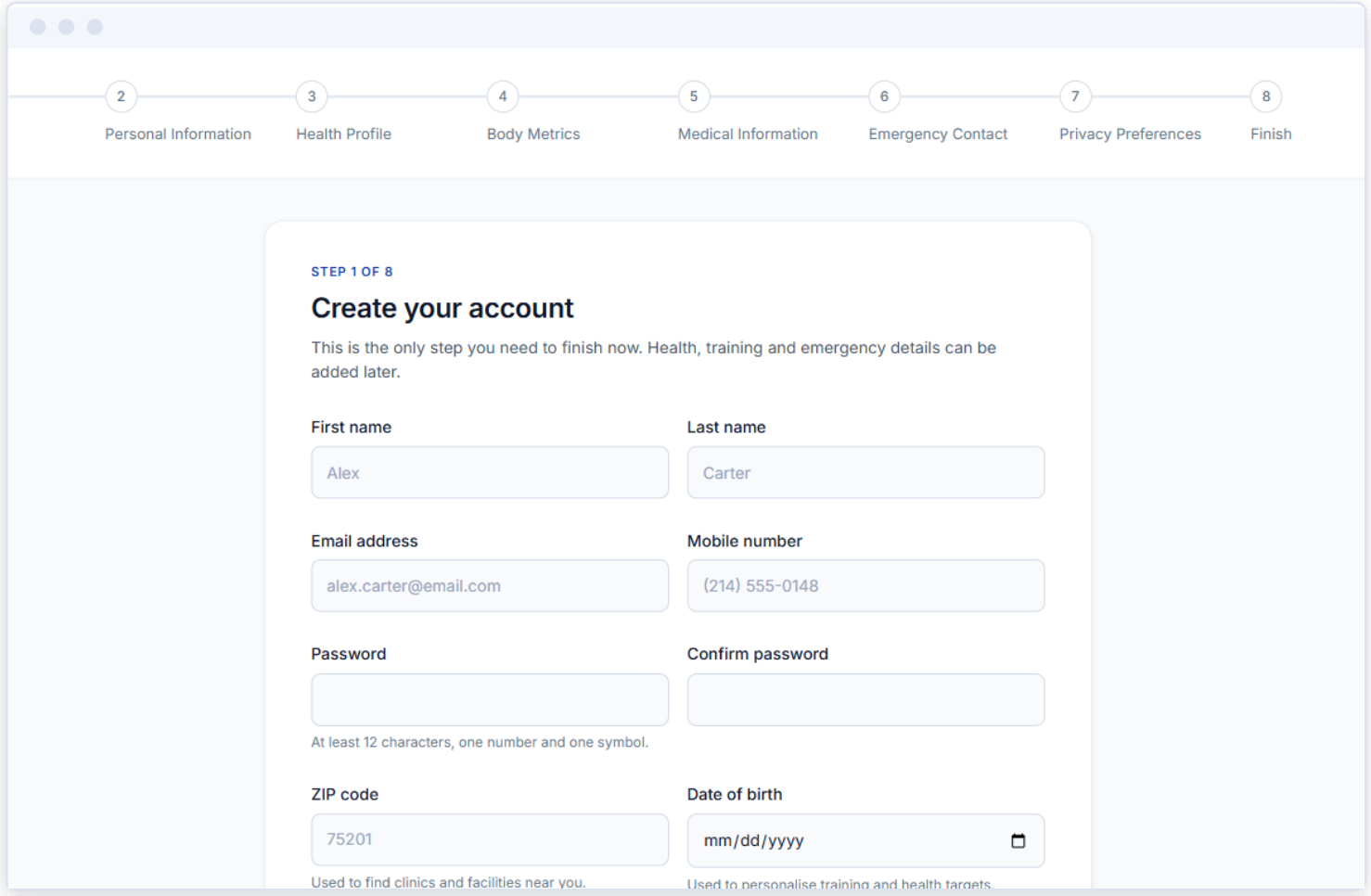
Sign-up, setup and control over personal data

What it does

Email or Google sign-up, an eight step setup, two step verification, data export and account deletion.

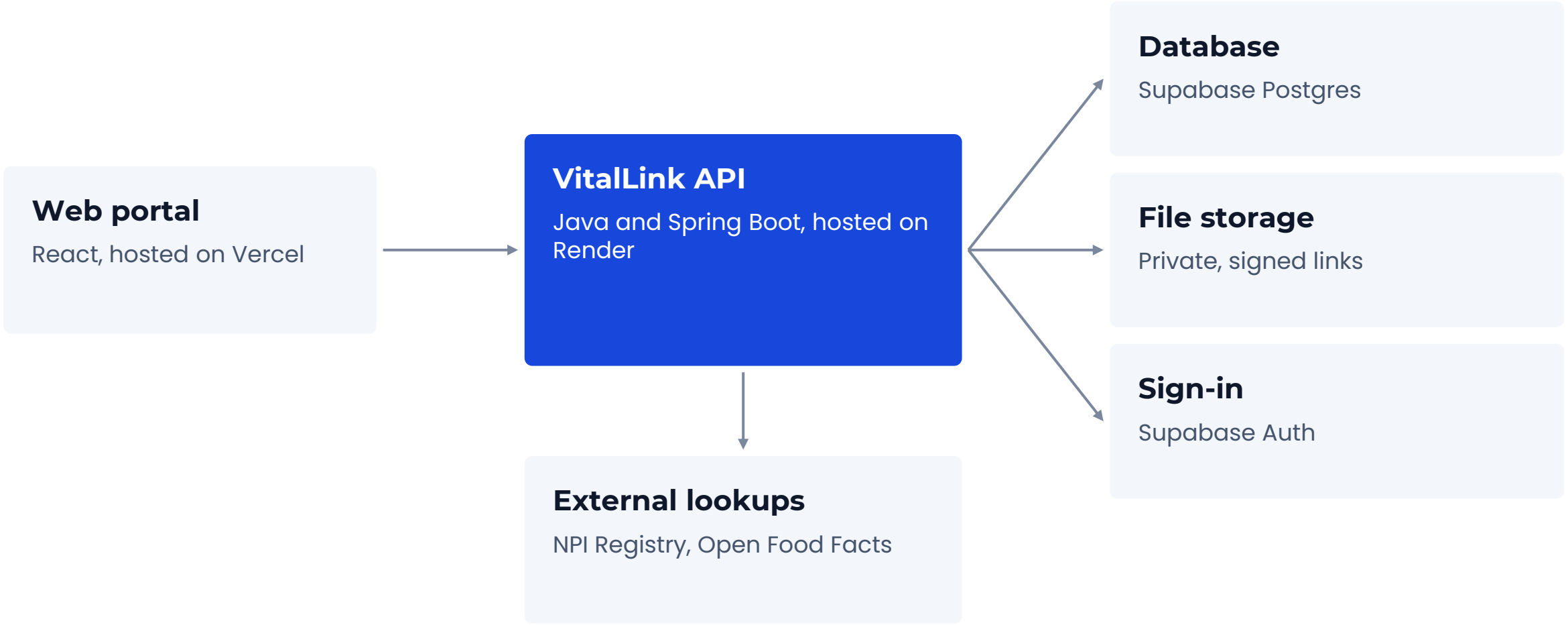
How it works

- The account records whether setup is finished, so Google sign-ups answer the same questions.
- With two step on, the API refuses a session that only passed the password.
- Deleting an account removes its data and ends its shares.



System Architecture

All rules live in one API, in front of the database and files



Technology Stack

The tools used for each part of the system

Part	Tools
Web portal	React 19, TypeScript, Vite 5, React Router 6, CSS Modules
API	Java 25, Spring Boot 4.1, Maven, OpenAPI with Swagger UI
Database	PostgreSQL on Supabase, plain SQL, Flyway (38 tables, 20 migrations)
Sign-in and files	Supabase Auth and Storage, Nimbus JOSE for verifying tokens
Hosting	Vercel for the site, Render with Docker for the API, Brevo for email
Automation	GitHub Actions for tests, builds and the nightly job
Design	Figma, with colors, spacing and type generated from its variables

Testing

Hands-on testing of the live site, supported by automated checks

Manual testing

- Sign-up with email and Google
- Setup and logging activity
- Apple Health import
- Document upload and sharing
- Revoking a share and reading the access log

Automated checks

286

API tests on a real database

102

frontend tests

47

step check of the running system

Issues found and fixed

API freeze under load

Many saves at once blocked for 10 seconds. Concurrent requests are now limited.

Apple Health imports

Large files failed. The size limit and parser were fixed, then tested with 2.5 million records.

Use of Generative AI

We used Claude Code, an AI coding tool, under the team's direction

Mostly produced by the AI tool

- Frontend and backend code
- Automated tests
- Documentation
- First drafts of new screens in Figma

The team's responsibility

- Requirements and project rules
- Product and design decisions
- Reviewing the work and testing it by hand
- Accounts, configuration and deployment

What we learned

AI output needs verification. One import test passed only because its sample file was small, and testing at a realistic size exposed the problem.

Live Demonstration

The working system, end to end

01 Sign up with Google and complete setup

02 Review the dashboard and My Health

03 Import Apple Health data

04 Log a workout and a meal

05 Add lab results and share them with a doctor

06 Search for a doctor and add a visit

07 Revoke access and review the access log

Live site

vital-link-flax.vercel.app

Limitations and Next Steps

What the system cannot do yet, and where it could go

Current limitations

- Apple Health data arrives by file import, not automatic sync
- The free server takes about a minute to wake after sitting idle
- A 500 MB free database limits how much data can be stored

Next steps

- Automatic daily updates from iPhone
- Usability testing with real users
- More database capacity and an always-on server if usage grows

Thank You

Questions and discussion

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